Digital VLSI Assignment: 4-bit Array Multiplier from RTL to GDSII

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SkyWater SKY130 `sky130_fd_sc_hd` standard-cell library

1 Introduction

This report implements a registered, unsigned 4-bit by 4-bit array multiplier and takes it from structural Verilog to a routed GDSII layout. The flow used Icarus Verilog for simulation, GTKWave for waveforms, Yosys for synthesis, OpenSTA for timing analysis, and LibreLane for physical design. Every number below was regenerated with the current IIC-OSIC-TOOLS image (`hpretl/iic-osic-tools:latest`, digest `sha256:306585de35d1`); the tool versions are listed in Section 7. All generated artifacts, scripts, logs and the final GDSII are included in the repository.

2 Design and verification (Part A)

The multiplier registers `a[3:0]` and `b[3:0]`, forms the 16 partial-product bits $pp_i[j] = a[j] \,\&\, b[i]$, and adds rows 1–3 with exactly 12 instances of `full_adder`, following the connection table of the assignment. Row 0 contributes `p[0]`; the lowest sum of each later row is `p[1]`, `p[2]` and `p[3]`, and row 3 also produces `p[7:4]`. The combinational result is captured in the 8-bit output register, so there are 16 AND gates, 12 full adders and 16 flip-flops, all with asynchronous active-low reset. An input applied before edge 1 is captured at edge 1 and its product is captured at edge 2, giving a latency of 2 cycles and a throughput of 1 result per cycle.

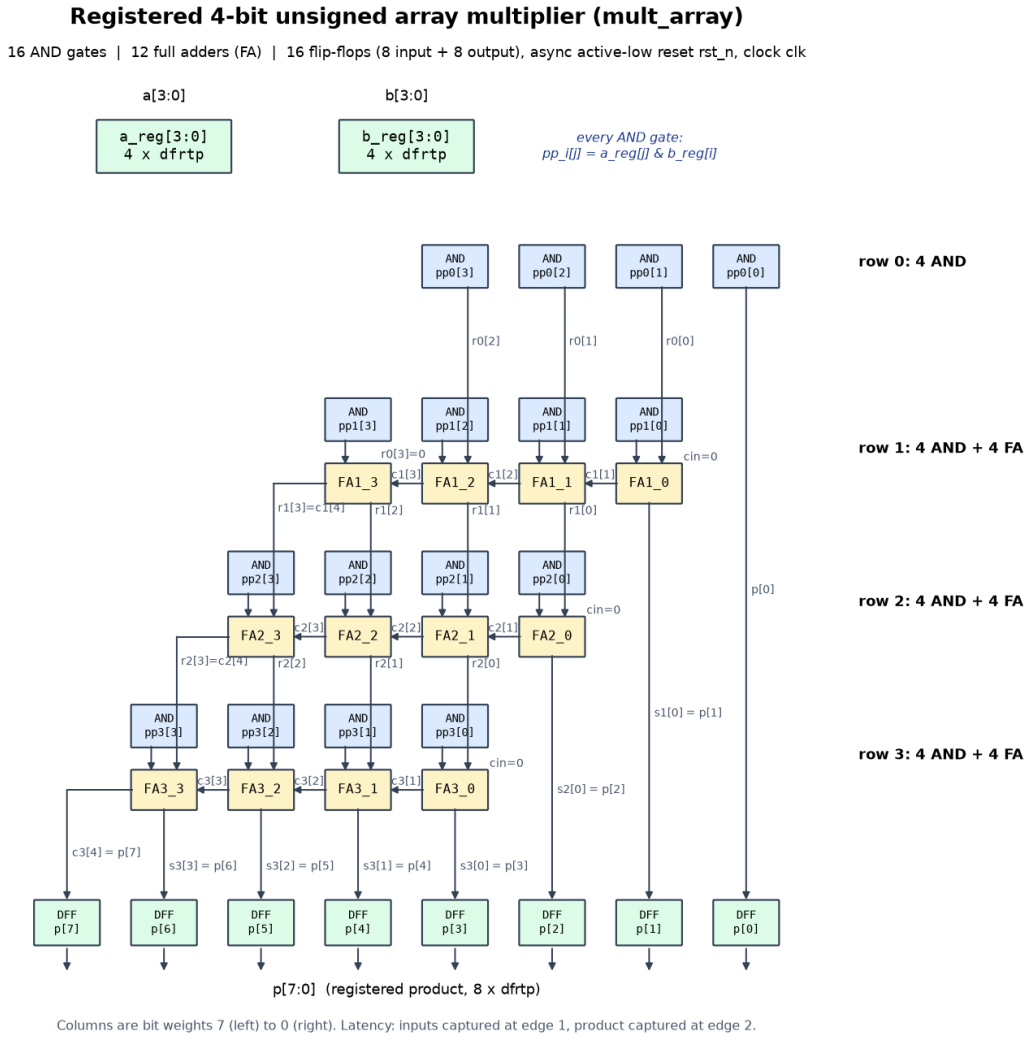


Figure 1: Block diagram: input registers, the 16 AND gates and 12 full adders arranged by bit weight, and the 8-bit output register.

The self-checking testbench `tb/tb_mult.v` uses a 10 ns clock, holds `rst_n` low for the first 2 cycles, applies all 256 input pairs, waits for the 2-cycle latency and compares `p` with `a * b`, printing every mismatch and writing `mult.vcd`.

It samples `p` on the falling edge after the capture edge, which keeps the comparison clear of the flip-flop clock-to-Q update in both RTL and gate-level simulation. RTL simulation reported `PASS: 256 vectors checked, 0 errors`. To prove the checker detects errors, `sum = a ^ b ^ cin` was temporarily changed to `sum = a ^ b`; the test then reported `FAIL: 256 vectors checked, 95 errors`, after which the correct RTL was restored.

For the required personal example we used Jainil Bavishi's roll number 2023102045: the last digit is 5, so $A = 5+5 = 10$ and $B = 7$. Worked by hand in the partial-product layout:

```
        1 0 1 0      A = 10
      x 0 1 1 1      B = 7
   ---------------
        1 0 1 0      pp0 = A AND b0 (b0 = 1)
      1 0 1 0        pp1 = A AND b1 (b1 = 1), shifted 1
    1 0 1 0          pp2 = A AND b2 (b2 = 1), shifted 2
  0 0 0 0            pp3 = A AND b3 (b3 = 0), shifted 3
   ---------------
0 1 0 0 0 1 1 0      P = 70
```

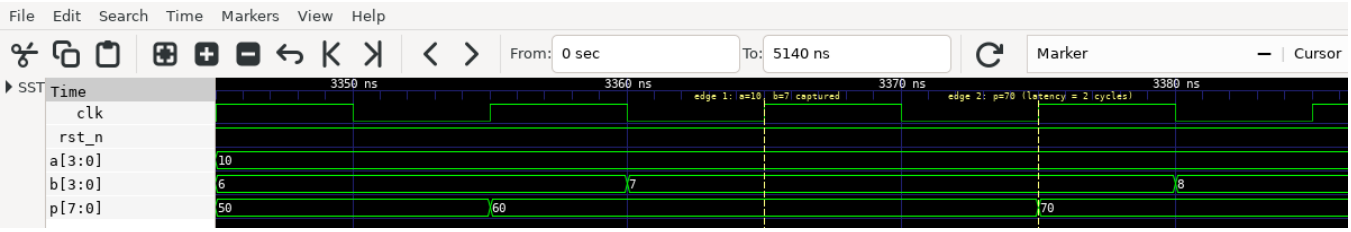


Figure 2: GTKWave capture of `mult.vcd` for $A = 10$, $B = 7$. The operands are applied at 3360 ns, captured at edge 1 (marker A, 3365 ns), and `p=70` is captured at edge 2 (marker B, 3375 ns): a latency of 2 clock cycles.

3 Synthesis (Part B)

Yosys flattened the hierarchy, mapped registers with `dfflibmap`, mapped combinational logic with `abc`, and wrote the SKY130 gate-level netlist. Table 1 is taken from `synth/mult_array_stat.rpt`. The netlist was simulated with the assignment's exact command (`-DFUNCTIONAL -DUNIT_DELAY=#1`) and the same testbench.

Table 1: Yosys synthesis results in the SKY130 standard-cell library.

Metric	Measured value
Total number of cells	61 cells
Number of flip-flops	16 `sky130_fd_sc_hd__dfrtp_1` cells
Total cell area	756.976 μm^2
Flip-flop area	$16 \times 25.024 = 400.384$ μm^2
Flip-flop area (% of total)	52.89%
Three most-used cell types	`dfrtp_1`: 16; `nand2_1`: 7; `maj3_1` and `and2_0`: 4 each
Netlist simulation	PASS: 256 vectors checked, 0 errors

4 Static timing analysis (Part C)

OpenSTA used the typical `tt_025C_1v80` Liberty corner, a 1.00 ns input delay and a 1.00 ns output delay. At the initial 10.00 ns period the worst setup slack was 7.21 ns, so $T_{min} = 10.00 - 7.21 = 2.79$ ns and $F_{max} = 1000/2.79 = 358.42$ MHz. Re-running at 2.79 ns gave a worst slack of +0.003 ns (0.00 ns at two decimals), confirming the result; at 2.78 ns the slack is −0.01 ns.

The critical path, from `sta/mult_array_timing.rpt`, is `dfrtp_1` (`a_reg[1]`) → `clkinv_1` → `o311ai_0` → `a21oi_1` → `maj3_1` → `xnor3_1` → `maj3_1` → `maj3_1` → `xor3_1` → `dfrtp_1` (`p[6]`), with a data arrival time of 2.652 ns. Its end point is the sum of `fa_r3_3`, the last adder of row 3. Five of the eight combinational cells are recognisable full-adder functions: the three `maj3` cells are full-adder carry outputs and the `xnor3`/`xor3` cells are full-adder sum outputs. The first three cells (`clkinv`, `o311ai`, `a21oi`) implement partial-product AND gates merged with the simplified row-1 adder logic, whose carry-in and top input are constant 0. The path therefore has 8 combinational

Table 2: OpenSTA maximum-delay timing results.

Metric	Measured value
Worst slack at $T = 10$ ns	7.21 ns
T_{min}	2.79 ns
F_{max}	358.42 MHz
Slack at $T = T_{min}$	+0.003 ns
Critical-path start point	`_91_/CLK` (input register `a_reg[1]`)
Critical-path end point	`_88_/D` (output register `p[6]`)
Number of cells on the critical path	10 (8 combinational + 2 flip-flops)
Latency $= 2T_{min}$	5.58 ns
Throughput	358.42 million results/s

cell delays, the same count as the $2N = 8$ full-adder-delay estimate. The match is approximate, because each cell implements only the carry or the sum of one adder (or a merged AND), and synthesis does not keep one library cell per full adder.

5 Physical design (Part D)

The configuration uses `FP_SIZING = "absolute"` and a fixed `DIE_AREA = [0, 0, 150, 150]`. The required first period is $1.2 \times 2.79 = 3.348$ ns, rounded up to 3.50 ns. Table 3 lists every attempt; the setup slack is LibreLane's `timing__setup__ws`, the worst over all signoff corners. The period was increased by 0.50 ns after each failing run. In every run the limiting corner was the slow `max_ss_100C_1v60` corner, while the typical-corner slack was positive in every run (+0.431 ns already at 3.50 ns). Slack did not improve steadily with the period: from 6.00 ns to 7.50 ns it stayed between −0.04 and −0.21 ns. LibreLane re-runs synthesis, placement and timing repair for each target period, and the buffers it puts on the critical net change with the period: up to 6.00 ns they were fast `buf_4` rebuffers, at 6.50 and 7.00 ns a slow `clkdlybuf4s25` delay cell was used as a max-capacitance buffer, and from 7.50 ns a second one was added as a wire buffer. Together the two delay cells add about 1.24 ns at the slow corner. The 7.50 ns and 8.00 ns runs produced the same critical path (arrival 7.64 ns), so the extra 0.50 ns at 8.00 ns turned −0.138 ns into +0.362 ns. This is why 10 runs were needed, compared with 7 when this design was first implemented with an older tool image.

Table 3: LibreLane timing-closure attempts.

Run	`CLOCK_PERIOD` (ns)	WNS, all corners (ns)	WNS, nom tt (ns)	Setup violations	DRC / LVS	Result
1	3.50	−2.546	+0.431	15	0 / 0	setup violations
2	4.00	−2.027	+0.929	15	0 / 0	setup violations
3	4.50	−1.527	+1.429	12	0 / 0	setup violations
4	5.00	−1.027	+1.929	9	0 / 0	setup violations
5	5.50	−0.527	+2.429	9	0 / 0	setup violations
6	6.00	−0.037	+2.911	1	0 / 0	setup violations
7	6.50	−0.205	+3.111	6	0 / 0	setup violations
8	7.00	−0.116	+3.526	5	0 / 0	setup violations
9	7.50	−0.138	+3.722	5	0 / 0	setup violations
10	8.00	+0.362	+4.222	0	0 / 0	pass

Place and route grew the design from 61 cells after synthesis to 1,884 instances. Only 86 of them are functional cells: 62 logic and flip-flop cells (LibreLane re-synthesizes the RTL with its own Yosys script, giving one more logic cell than Part B), 21 timing-repair buffers and 3 clock buffers. The other 1,798 instances are physical-only cells with no logic function: 1,299 decap cells, 254 filler cells and 245 tap cells.

Table 4: Final LibreLane physical-design results.

Metric	Measured value
CLOCK_PERIOD used	8.00 ns
Number of runs needed	10
Core area / die area	17,759.5 μm^2 / 22,500 μm^2
Cells after synthesis (Part B)	61
Cells after place and route	1,884 instances (86 excluding fillers and taps)
Kinds of cells added	1,299 decap + 254 filler cells; 245 tap cells; 21 timing-repair buffers; 3 clock buffers
Wirelength	2,518 μm
Setup slack	0.3618 ns
Post-route $F_{max} = 1000/(T - \text{slack})$	130.92 MHz
F_{max} from Part C	358.42 MHz
DRC errors (Magic / KLayout)	0 / 0
LVS errors	0

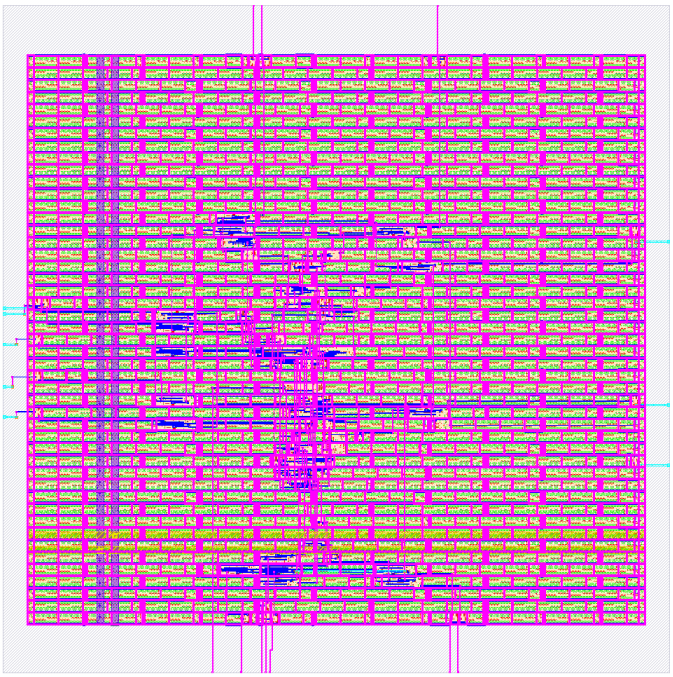


Figure 3: Final 150 μm by 150 μm GDSII layout in KLayout, whole-die view.

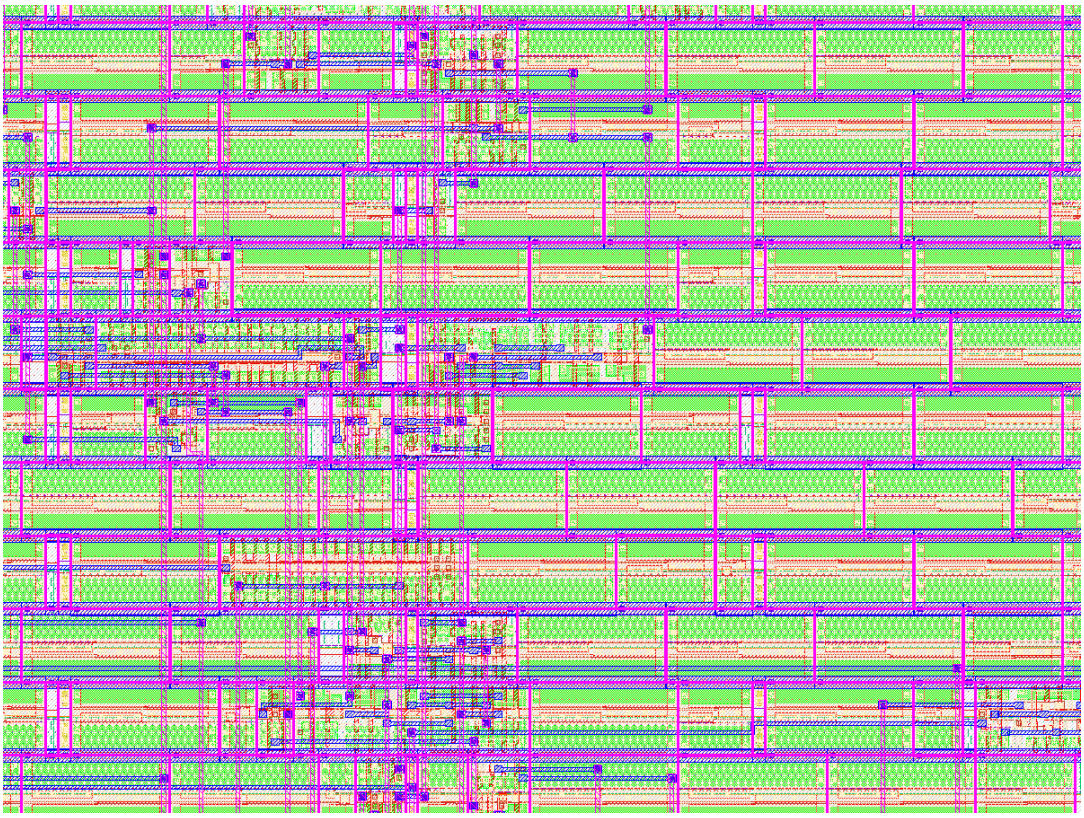


Figure 4: KLayout close-up of the final layout showing standard-cell rows, power rails and routed metal.

6 Answers to questions

1. Synthesis produced $61 - 16 = 45$ logic cells. This is not simply $16 + 12 = 28$ because a full adder is not mapped to one cell: `abc` maps each carry to a `maj3` or AOI/OAI gate and each sum to XOR/XNOR gates, so one full adder becomes about two to three cells. Many AND gates are not separate cells either; they are merged into NAND, AOI and OAI gates that drive the adders, and the constant-0 inputs of the first adders are simplified away.

2. The critical path starts at input register `a_reg[1]` (`_91_`) and ends at output register `p[6]` (`_88_`, the sum of `fa_r3_3`). It has 10 cells: the two flip-flops and 8 combinational cells, of which 5 (`maj3` ×3, `xnor3`, `xor3`) are full-adder carry/sum functions and 3 are merged AND/row-1 logic. Its 8 combinational cell delays match the $2N = 8$ full-adder-delay estimate for $N = 4$ in count, although each cell is only part of a full adder (its carry or its sum), so the agreement is approximate.

3. Flip-flops occupy $400.384/756.976 \times 100 = 52.89\%$ of the synthesized cell area. A resettable flip-flop is a large cell: one `dfrtp_1` is 25.02 μm^2, against 3.75 μm^2 for a `nand2_1` and 10.01 μm^2 for a `maj3_1`. The design needs 16 of them, while a 4-bit array has only 45 mostly small logic cells. The register overhead is fixed by the I/O width, so in a tiny datapath it dominates the area.

4. The post-route F_{max} of 130.92 MHz is lower than the 358.42 MHz from Part C because Part C used ideal wires, an ideal clock and the typical corner only. Post-route signoff checks every corner, and the limiting one is the slow `max_ss_100C_1v60` corner (the typical-corner slack at 8.00 ns is +4.22 ns). The worst path is the same `a_reg[1]` → `p[6]` adder chain as in Part C, but it now arrives at 7.64 ns. At the slow corner every cell is about 2–2.5× slower (the flip-flop clock-to-Q alone grows from 0.40 to 0.99 ns), and two delay-cell repair buffers add 1.24 ns. The clock tree adds 0.57 ns of insertion delay at the launch flip-flop, but this is largely cancelled by 0.51 ns at the capture flip-flop. On the required side, 0.25 ns of clock uncertainty and a 0.29 ns slow-corner setup time are subtracted. Wire delay itself is small on a 150 μm die. The flow added 1,299 decap cells and 254 filler cells to fill the empty row space, continuing the power rails and wells; the decaps also add local supply decoupling capacitance. 245 tap cells tie the n-well and substrate to VPWR/VGND to prevent latch-up. 21 timing-repair buffers (17 `buf_2` and 4 `clkdlybuf4s` cells) fix maximum slew, capacitance and wire-length limits on long or heavily loaded nets. 3 `clkbuf_16` clock buffers form the clock tree that drives all 16 flip-flops with low skew.

5. DRC checks that the drawn layout obeys the foundry's geometry rules (minimum width, spacing, enclosure, density and so on), so it can be manufactured reliably. LVS extracts the transistors and connectivity from the layout and compares them with the intended netlist, so it confirms the layout is the circuit that was designed. Both must pass: a layout can meet every geometric rule but connect the wrong nets, or have the right connectivity but violate a rule that causes shorts or opens on silicon.

6. For an N-bit array, the number of AND gates is N^2 and the number of full adders is $N(N-1)$. At $N = 8$: $8^2 = 64$ AND gates and $8 \times 7 = 56$ full adders, against 16 and 12 at $N = 4$, so the full adders grow by $56/12 \approx 4.7\times$. The critical path grows from $2 \times 4 = 8$ to $2 \times 8 = 16$ full-adder delays, so the combinational delay roughly doubles. Scaling our measured path: its combinational part is $2.652 - 0.398 = 2.254$ ns, which doubles to about 4.51 ns; adding the 0.398 ns clock-to-Q and 0.135 ns setup time gives $T_{min} \approx 5.04$ ns, so F_{max} would drop from 358 MHz to about 198 MHz before layout effects.

7 Tool versions

IIC-OSIC-TOOLS image `hpretl/iic-osic-tools:latest` (`sha256:306585de35d1`); Icarus Verilog 14.0 (devel); GTKWave v4.0.0-prealpha (f6d60e0); Yosys 0.70 (28ba3cb92); OpenSTA 3.1.0 (80fbbb2945); LibreLane 3.1.0.dev3; KLayout 0.30.12; SKY130A PDK, `sky130_fd_sc_hd__tt_025C_1v80` Liberty for Parts B and C.

8 Conclusion

The structural RTL passed exhaustive RTL and gate-level simulation with the assignment's exact commands, and the fault-injection check confirmed that the testbench detects errors. Synthesis produced 61 cells (756.976 μm^2), STA gave $T_{min} = 2.79$ ns ($F_{max} = 358.42$ MHz), and LibreLane closed timing at an 8.00 ns clock after 10 runs with 0.3618 ns setup slack, 0 DRC errors and 0 LVS errors.